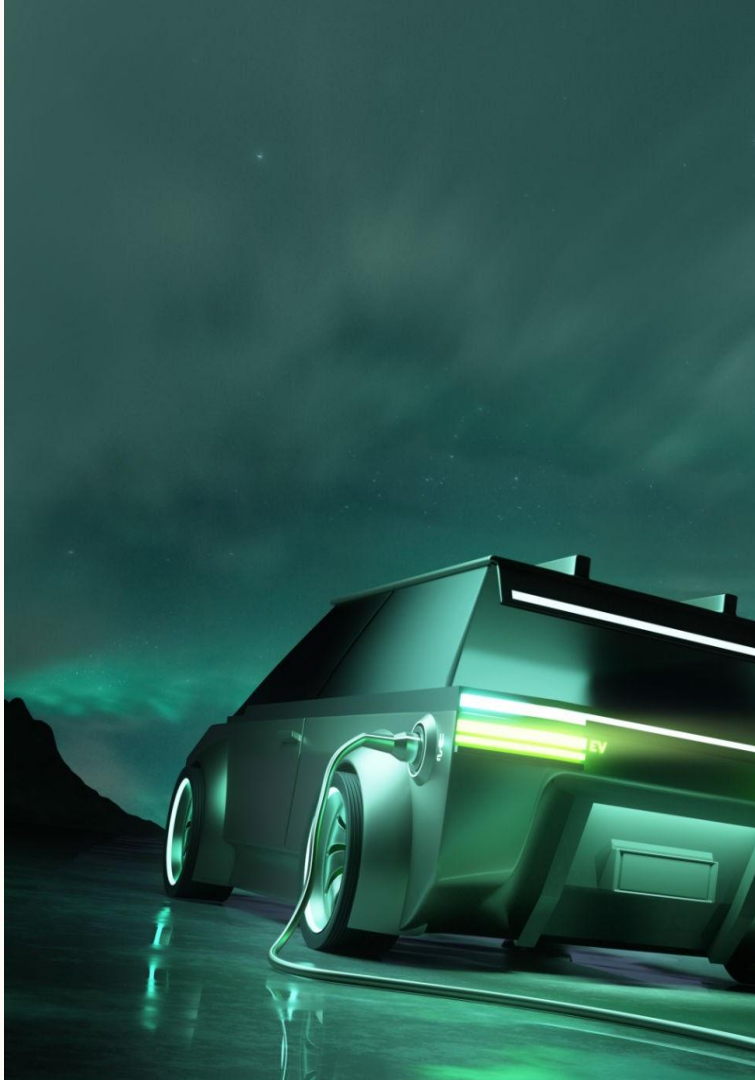
Electric Vehicle Battery Performance & Range Prediction

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Prajwal Auradkar (2320040084)



Data-Driven Analysis & Machine Learning Roadmap for EV Range Optimization

Objective

Predict real-world driving range from battery & vehicle telemetry

Scope

Data inspection, cleaning, multi-level EDA, ML modeling

Dataset

2,000 operational records · 17 real-time parameters (SOC, Speed, Temp, Power, Voltage, Driving Style)

Dataset Health & Preprocessing Standards

Concise protocols for reproducible, robust preprocessing

01

Structural Audit

Full summaries: .head, .tail, .info, .describe — schema & types verified

02

Imputation Strategy

Symmetric (|skew| ≤ 0.5) → Mean (e.g., Battery Capacity = 65.11 kWh, Ambient Temp = 25.02°C)

03

Skewed Features

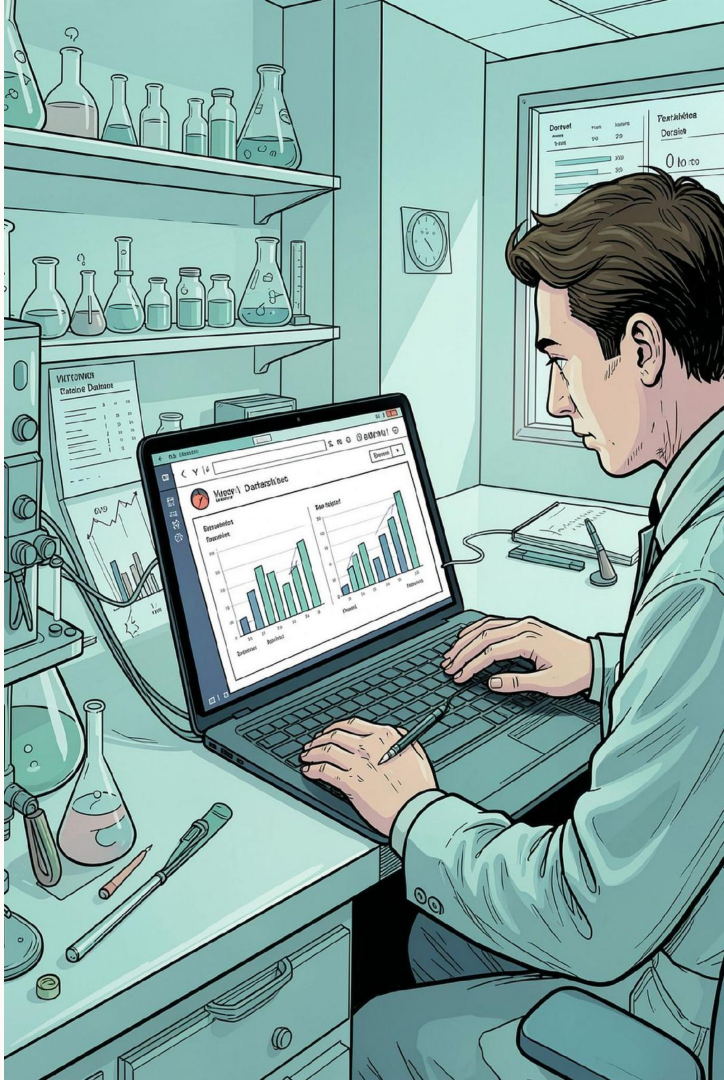
|skew| > 0.5 → Median (e.g., Charge Cycles = 1002, Storage Time = 6.0 months)

04

Outlier Diagnostics

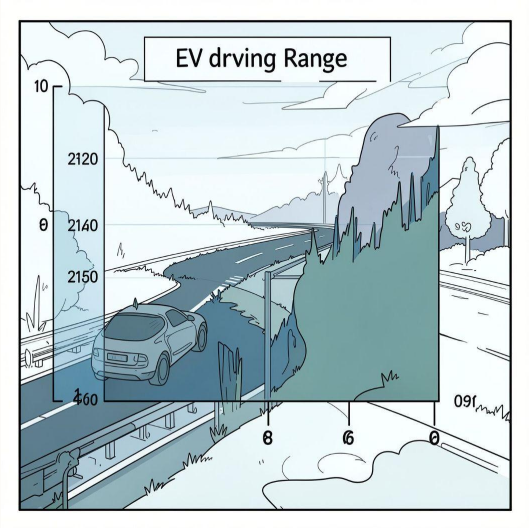
IQR bounds (Q1 − 1.5·IQR, Q3 + 1.5·IQR) to isolate extreme events vs noise

Use domain thresholds (voltage, current, temperature) to flag sensor faults before imputation.



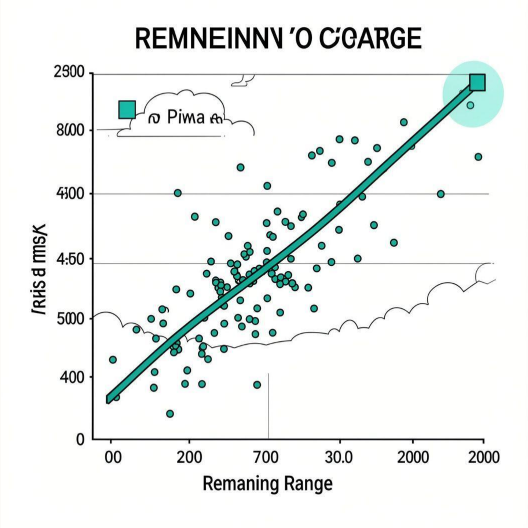
Multi-Level Exploratory Data Analysis (EDA)

Key findings across uni-, bi-, and multi-variate views



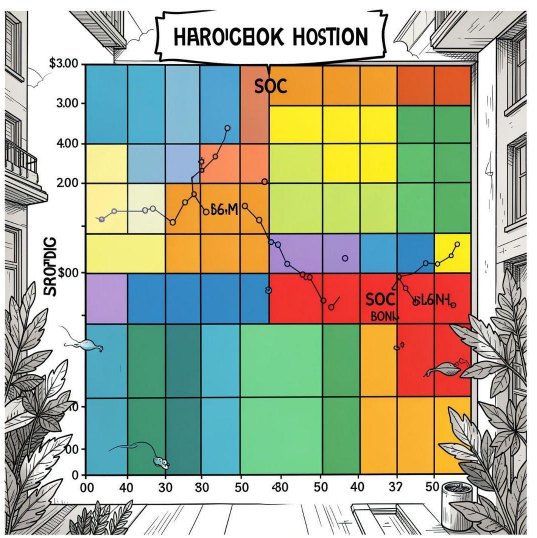
Univariate

Remaining Range — mean ≈ 105.9 km. Distribution by Driving Style: Eco / Normal / Aggressive balanced.



Bivariate

SOC ↔ Remaining Range: strong linear dependence. Eco driving yields higher usable range at identical SOC.



Multivariate

Correlation heatmap: SOC r = 0.97 (primary predictor). Aerodynamic drag (high speed) r = -0.12 (negative effect).

Advanced Factor Interactions & Diagnostic EDA



SOC Thresholds

Low SOC (<30%) ⇒ Remaining range <50 km. Critical cutoff zones for safe operation.



Electrical Spread

System voltage centered ≈ 360V (300–420V). Current spikes 10A–200A correlated with acceleration events.



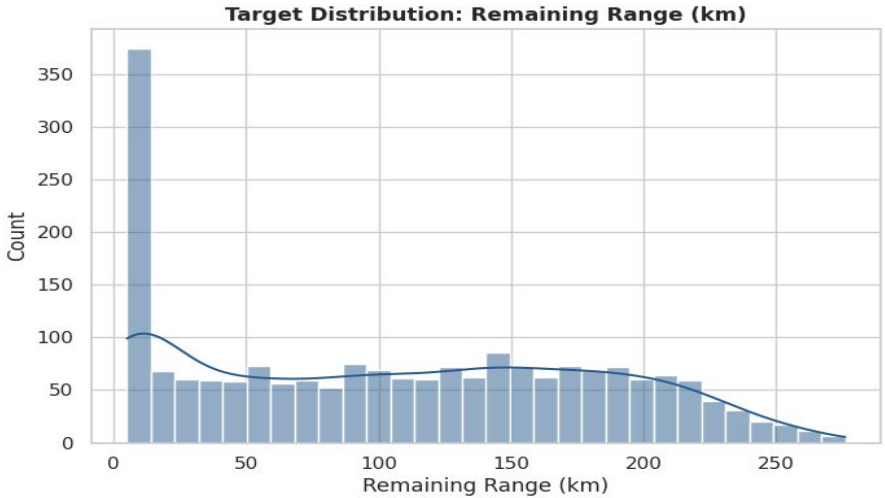
Thermal Trends

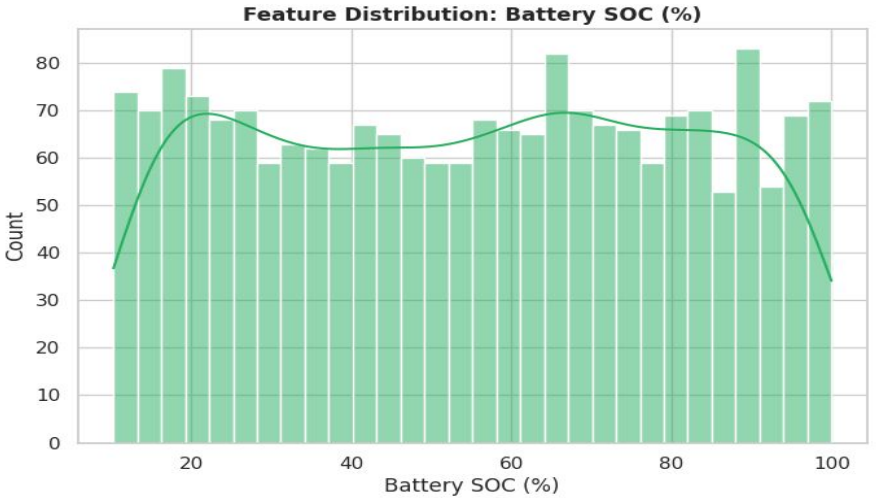
Higher ambient temps → measurable range degradation (regression slope small but consistent).

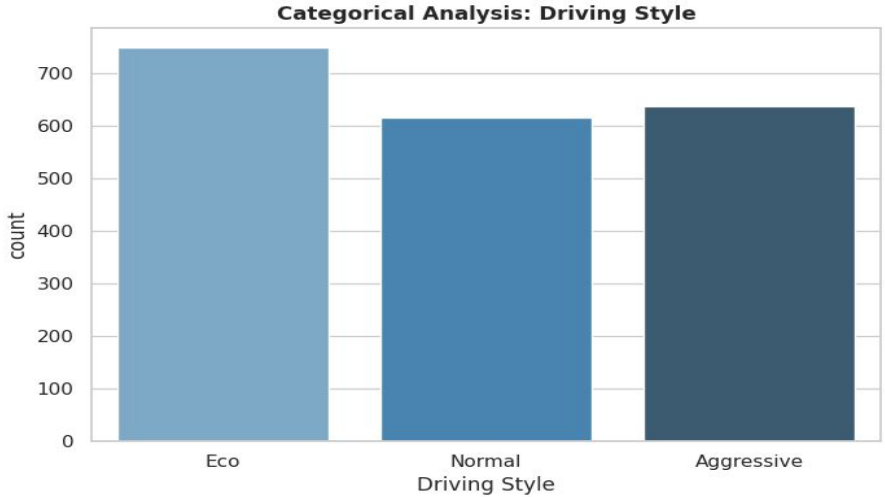


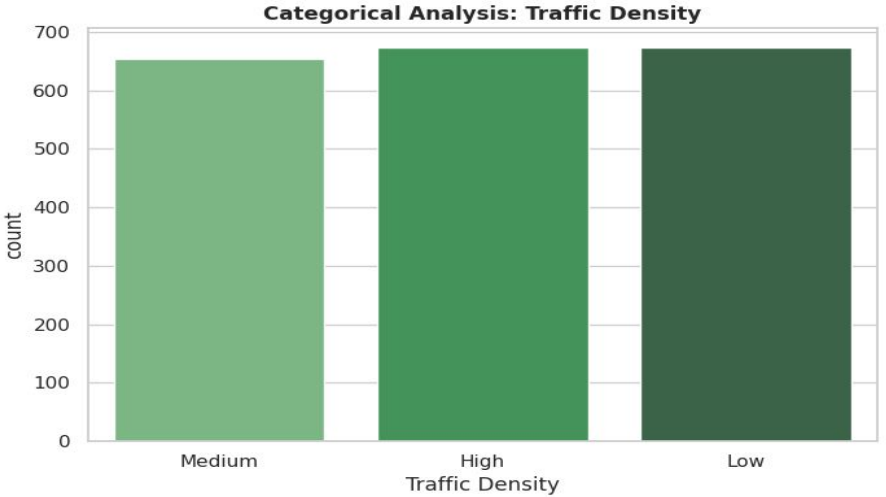
Feature Interactions

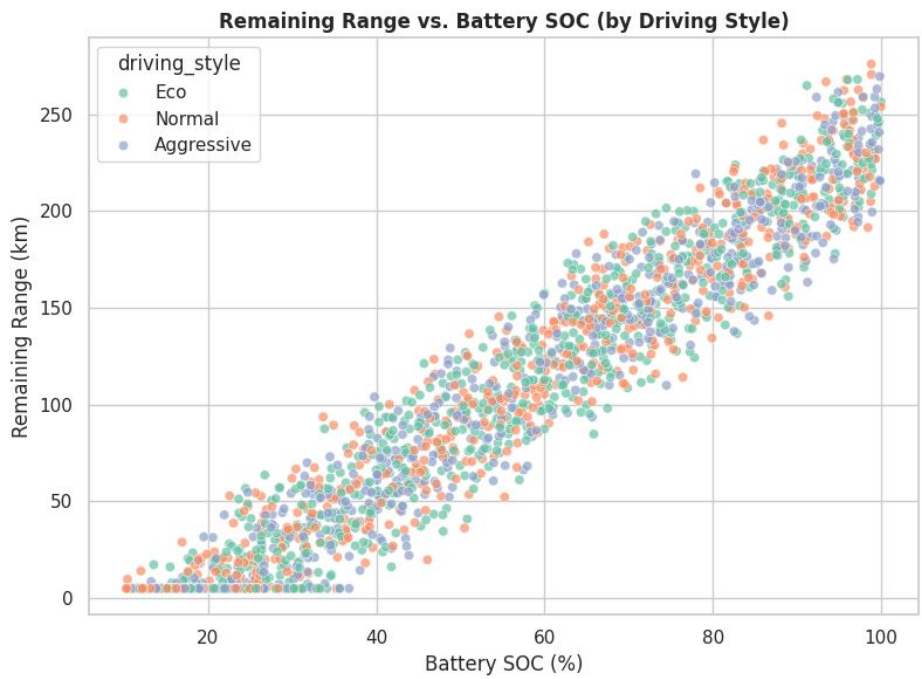
Pairwise checks show environmental variables largely independent → low multicollinearity for regression models.


Target Distribution: Remaining Range (km)
Count
Remaining Range (km)
0
50
100
150
200
250
0
50
100
150
200
250
300
350

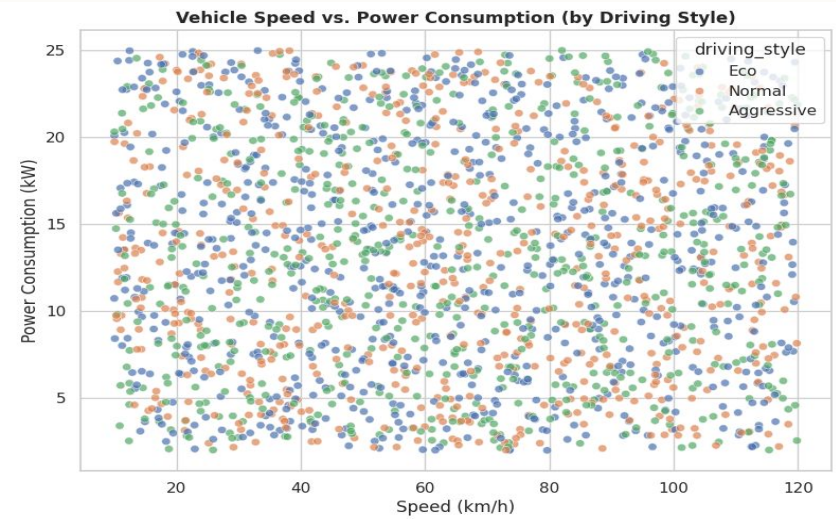

Feature Distribution: Battery SOC (%)
Count
Battery SOC (%)
20
40
60
80
100
0
10
20
30
40
50
60
70
80

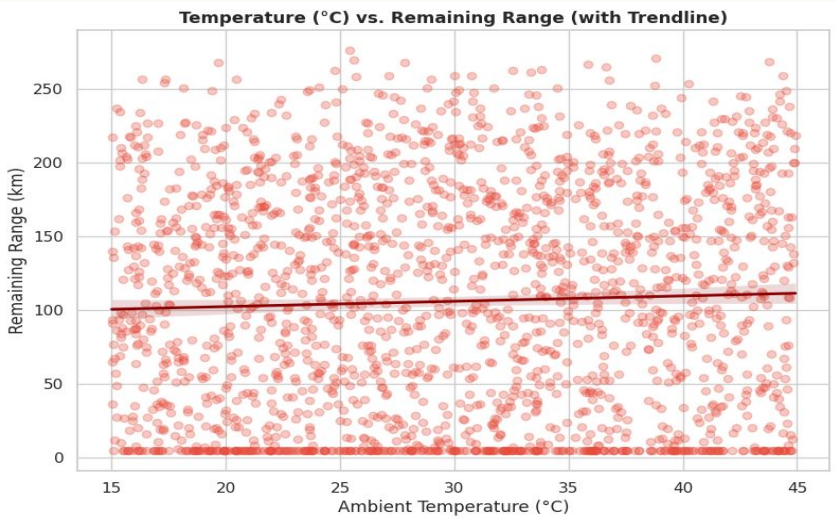

Categorical Analysis: Driving Style
count
Eco
Normal
Aggressive
Driving Style
0
100
200
300
400
500
600
700

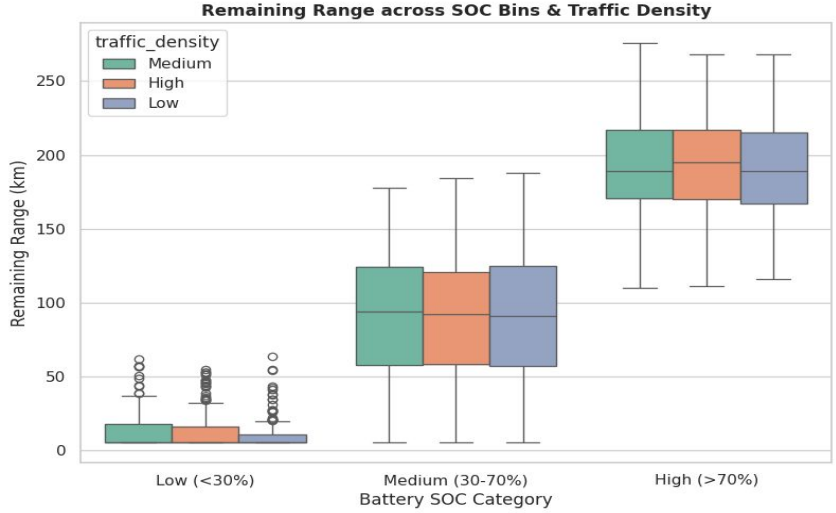

Categorical Analysis: Traffic Density
count
Medium
High
Low
Traffic Density
0
100
200
300
400
500
600
700

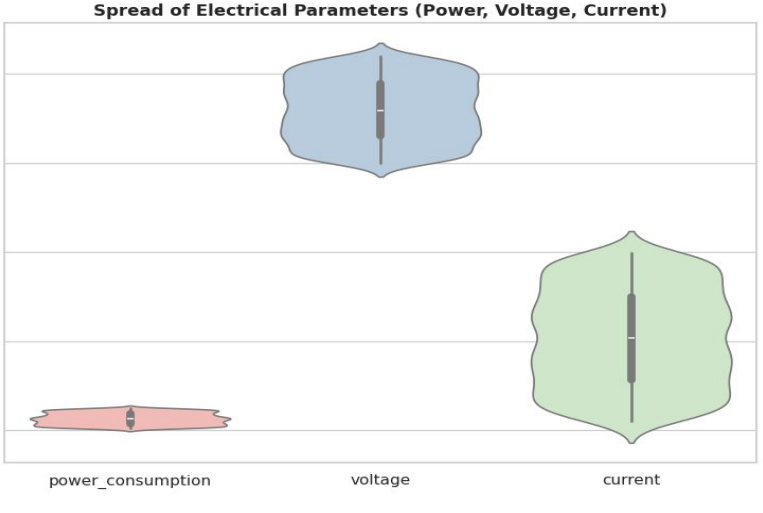

Remaining Range vs. Battery SOC (by Driving Style)
driving_style
Eco
Normal
Aggressive
Remaining Range (km)
0
50
100
150
200
250
20
40
60
80
100
Battery SOC (%)

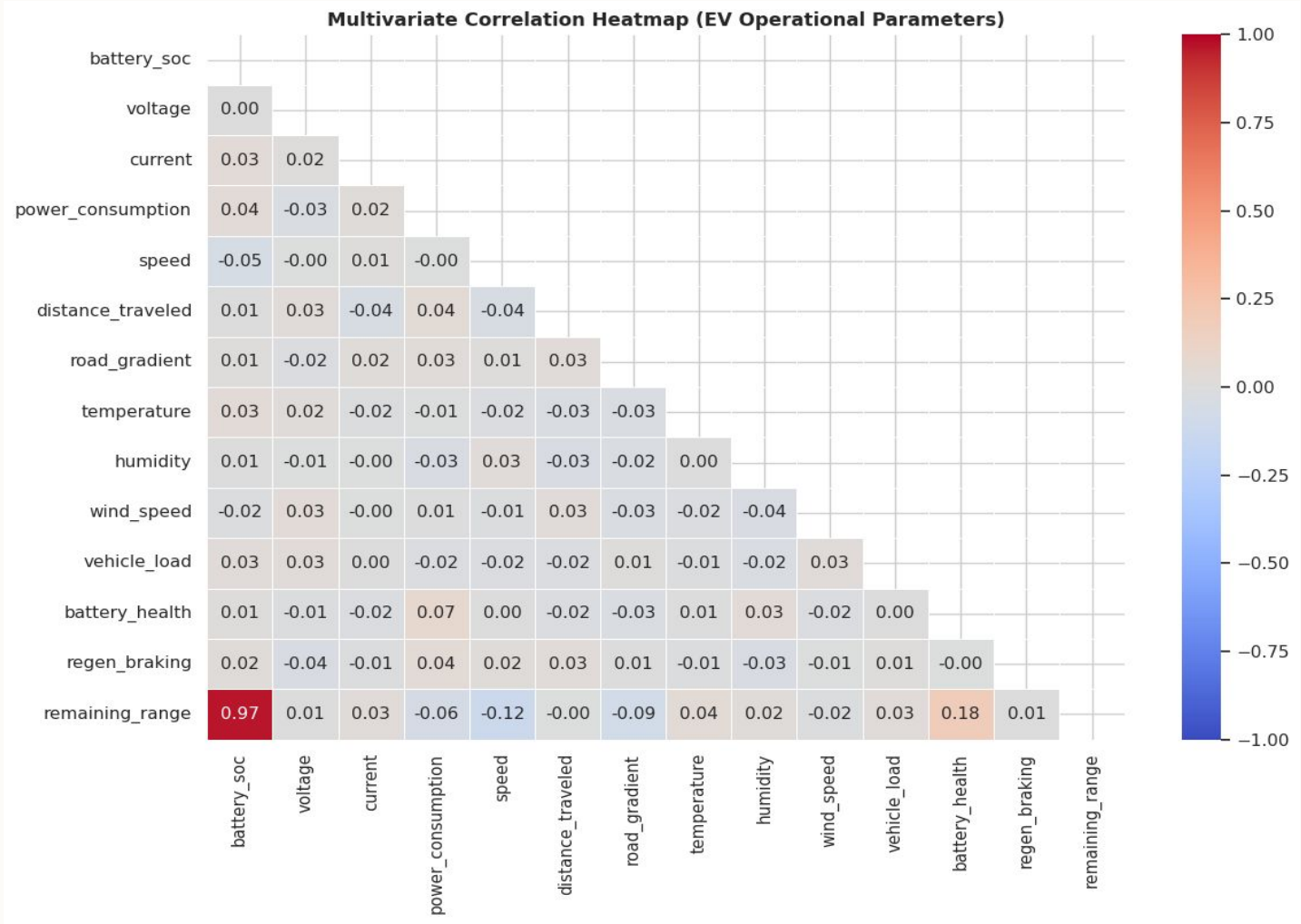

Remaining Range Across Driving Styles & Traffic Density
traffic_density
Medium
High
Low
Remaining Range (km)
0
50
100
150
200
250
Eco
Normal
Aggressive
Driving Style


Vehicle Speed vs. Power Consumption (by Driving Style)
driving_style
Eco
Normal
Aggressive
Power Consumption (kW)
Speed (km/h)


Temperature (°C) vs. Remaining Range (with Trendline)
Remaining Range (km)
Ambient Temperature (°C)

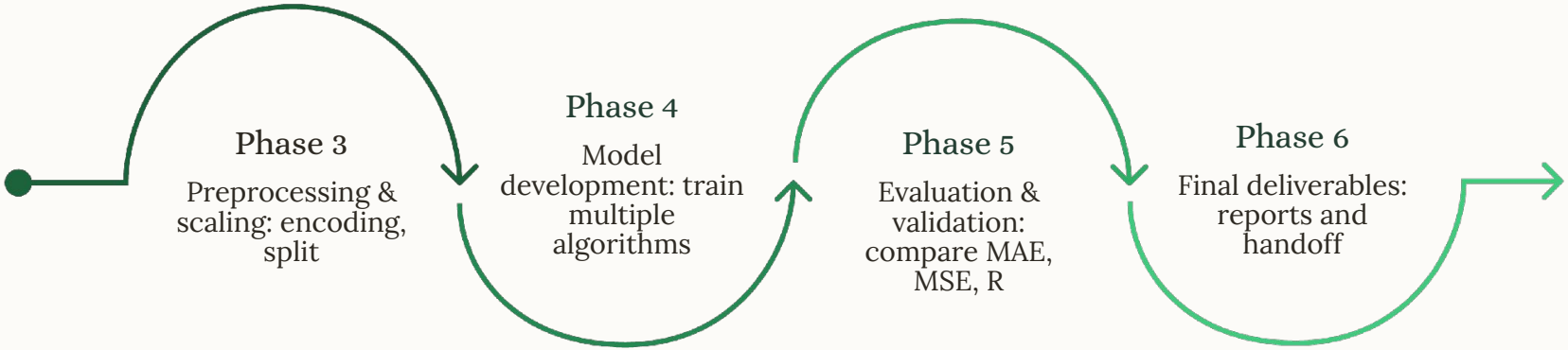

Remaining Range across SOC Bins & Traffic Density
traffic_density
Medium
High
Low
Remaining Range (km)
Low (<30%)
Medium (30-70%)
High (>70%)
Battery SOC Category


Spread of Electrical Parameters (Power, Voltage, Current)
power_consumption
voltage
current


Multivariate Correlation Heatmap (EV Operational Parameters)
battery_soc
voltage 0.00
current 0.03 0.02
power_consumption 0.04 -0.03 0.02
speed -0.05 -0.00 0.01 -0.00
distance_traveled 0.01 0.03 -0.04 0.04 -0.04
road_gradient 0.01 -0.02 0.02 0.03 0.01 0.03
temperature 0.03 0.02 -0.02 -0.01 -0.02 -0.03 -0.03
humidity 0.01 -0.01 -0.00 -0.03 0.03 -0.03 -0.02 0.00
wind_speed -0.02 0.03 -0.00 0.01 -0.01 0.03 -0.03 -0.02 -0.04
vehicle_load 0.03 0.03 0.00 -0.02 -0.02 -0.02 0.01 -0.01 -0.02 0.03
battery_health 0.01 -0.01 -0.02 0.07 0.00 -0.02 -0.03 0.01 0.03 -0.02 0.00
regen_braking 0.02 -0.04 -0.01 0.04 0.02 0.03 0.01 -0.01 -0.03 -0.01 0.01 -0.00
remaining_range 0.97 0.01 0.03 -0.06 -0.12 -0.00 -0.09 0.04 0.02 -0.02 0.03 0.18 0.01
battery_soc
voltage
current
power_consumption
speed
distance_traveled
road_gradient
temperature
humidity
wind_speed
vehicle_load
battery_health
regen_braking
remaining_range
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Path to Project Completion

Phased roadmap: preprocessing → modeling → evaluation → delivery



1

Phase 3

One-hot encoding, 80/20 split, standard scaling

2

Phase 4

Train: Linear Reg, Decision Tree, Random Forest, Gradient Boosting

3

Phase 5

Compare MAE, MSE, RMSE, R^2 → select best model

4

Phase 6

Deliverables: final report, viva prep, GitHub repo & reproducible pipelines

Next steps: finalize feature engineering (SOC curves, temperature interaction terms) and baseline model benchmarks within two sprints.